

MCSA-P03 — PROJECT EVIDENCE

Remote Employee Access

Prepared by:
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Lab Environment: VMware Virtual Environment
Operating System: Microsoft Windows Server 2016
Domain: Aster.com

Project Overview

Project Objective

The Project 3 workbook defines a Remote Employee Access service for the fictional Aster Business Services environment. The project implements RRAS remote-access VPN, L2TP over IPsec, routing and return paths, dial-in authorization, DNS over VPN, RDP with Network Level Authentication, and scoped firewall controls.

Business Scenario

The workbook scenario requires Anita to reach the internal HR folder from a simulated home network. The lab must prove that internal access is unavailable before VPN, available through an authorised tunnel, and denied again after remote access is revoked.

Infrastructure

System	Role	Network	Address
AST-DC01	AD DS / DNS / DHCP	VMnet2	192.168.50.10/24
AST-FS01	File server / SMB	VMnet2	192.168.50.20/24
AST-BK01	Backup target	VMnet2	192.168.50.30/24
AST-CL01	Test client	VMnet2	DHCP 192.168.50.100–149
AST-VPN01	Remote-access VPN server	VMnet2 + VMnet3	192.168.50.40/24 + 172.16.90.1/24
AST-REM01	Remote client	VMnet3	172.16.90.2/24

Technology areas

- Active Directory Domain Services and named user accounts
- DNS and DHCP foundation services
- SMB and NTFS controlled departmental access
- Group Policy security controls
- RRAS remote access VPN and L2TP/IPsec
- Routing and return paths
- Windows Firewall scoped rules
- RDP with Network Level Authentication

B01 — Prepare the isolated virtual network

.bit

The screenshot shows the 'Virtual Network Editor' window. At the top, there is a table listing virtual networks. 'VMnet2' is selected, highlighted in blue. Below the table are three buttons: 'Add Network...', 'Remove Network', and 'Rename Network...'. The 'VMnet Information' section contains several options: 'Bridged' (unselected), 'NAT' (unselected), and 'Host-only' (selected). There are also checkboxes for 'Connect a host virtual adapter to this network' and 'Use local DHCP service to distribute IP address to VMs'. At the bottom, there are input fields for 'Subnet IP' and 'Subnet mask', and a 'Change Settings' button.

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	NAT	NAT	Connected	Enabled	192.168.238.0
VMnet2	Custom	-	-	-	192.168.50.0
VMnet3	Custom	-	-	-	172.16.90.0

Buttons: Add Network..., Remove Network, Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: [dropdown] Automatic Settings...

☐ NAT (shared host's IP address with VMs) NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☐ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet2

☐ Use local DHCP service to distribute IP address to VMs DHCP Settings...

Subnet IP: [192.168.50.0] Subnet mask: [255.255.255.0]

⚠ Administrator privileges are required to modify the network configuration. Change Settings

Buttons: Restore Defaults, Import..., Export..., OK, Cancel, Apply, Help

▼ Virtual Machine Details

State: Powered off

B02 — Build AST-DC01

The screenshot displays the Windows Server Manager interface. The left-hand navigation pane shows the 'Local Server' role selected. The main area is titled 'PROPERTIES For AST-DC01' and contains a grid of system information. Below this, an 'EVENTS' section shows a list of system events.

PROPERTIES For AST-DC01

Computer name	AST-DC01	Last installed updates	Never
Domain	Aster.com	Windows Update	Download updates only, using Windows Update
		Last checked for updates	Never
Windows Firewall	Public: On	Windows Defender	Real-Time Protection: On
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Ethernet0	192.168.50.10, IPv6 enabled	Product ID	00377-10000-00000-AA360 (activated)
Operating system version	Microsoft Windows Server 2016 Datacenter Evaluation	Processors	11th Gen Intel(R) Core(TM) i5-11320H @ 3.20GHz, 11th Gen Intel(R) Core(TM)
Hardware information	VMware, Inc. VMware7,1	Installed memory (RAM)	2 GB
		Total disk space	59.45 GB

EVENTS
All events | 25 total

Server Name	ID	Severity	Source	Log	Date and Time
AST-DC01	14553	Warning	Microsoft Windows-DfsSur	System	9/22/2026 11:11:04 AM

AST-DC01 Server Manager — local server properties and installed roles.

B03 — Build AST-FS01

The screenshot displays the Windows Server Manager interface for a local server named AST-FS01. The left-hand navigation pane includes links to the Dashboard, Local Server (which is currently selected), All Servers, and File and Storage Services. The main content area is titled 'PROPERTIES For AST-FS01' and contains a grid of system information. This information is organized into two sections: the top section covers updates, firewall, and security settings, while the bottom section details the operating system version, hardware, and processor information. Below the properties grid, there is an 'EVENTS' section showing a list of system events with columns for Server Name, ID, Severity, Source, Log, and Date and Time. The taskbar at the bottom of the screen shows the Start button, search icon, and several pinned applications, with the system clock indicating 11:13 AM on 9/22/2026.

PROPERTIES For AST-FS01			
Computer name	AST-FS01	Last installed updates	Never
Domain	Aster.com	Windows Update	Download updates only, using Windows Update
		Last checked for updates	Never
Windows Firewall	Public: On	Windows Defender	Real-Time Protection: On
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC+05:30) Sri Jayawardenepura
Ethernet0	192.168.50.20, IPv6 enabled	Product ID	00378-00000-00000-AA739 (activated)
Operating system version	Microsoft Windows Server 2016 Standard Evaluation	Processors	11th Gen Intel(R) Core(TM) i5-11320H @ 3.20GHz, 11th Gen Intel(R) Core(TM) i5-11320H @ 3.20GHz
Hardware information	VMware, Inc. VMware7,1	Installed memory (RAM)	2 GB
		Total disk space	79.32 GB

EVENTS					
All events 23 total					
Server Name	ID	Severity	Source	Log	Date and Time
AST-FS01	8016	Warning	Microsoft-Windows-DNS-Client-Events	System	9/22/2026 11:10:43 AM

AST-FS01 Server Manager — local server properties.

B04 — Build AST-BK01

Server Manager

Server Manager ▸ Local Server

Dashboard

Local Server

All Servers

File and Storage Services ▸

PROPERTIES

For AST-BK01

Computer name	AST-BK01	Last installed updates	Never
Domain	Aster.com	Windows Update	Download u
		Last checked for updates	Never
Windows Firewall	Domain: On	Windows Defender	Real-Time P
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC-08:00)
Ethernet0	192.168.50.30, IPv6 enabled	Product ID	Not activate
Operating system version	Microsoft Windows Server 2016 Standard Evaluation	Processors	11th Gen Int
Hardware information	VMware, Inc. VMware7,1	Installed memory (RAM)	2 GB
		Total disk space	119.32 GB

EVENTS

All events | 16 total

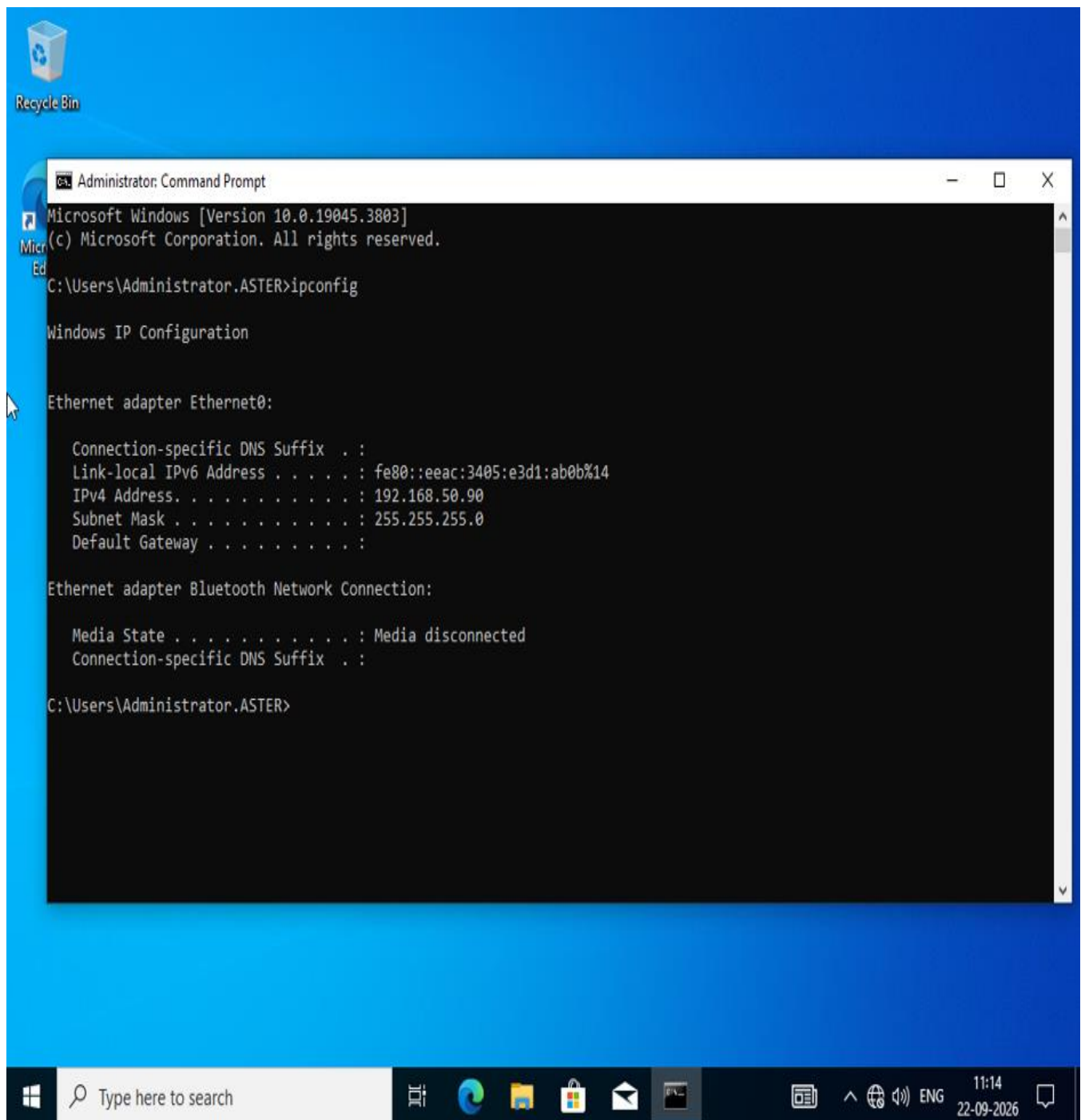
Filter

Server Name	ID	Severity	Source	Log	Date and Time
AST-BK01	8198	Error	Microsoft-Windows-Security-SPP	Application	9/21/2026 10:41:34 PM
AST-BK01	1014	Error	Microsoft-Windows-Security-SPP	Application	9/21/2026 10:41:34 PM
AST-BK01	8200	Error	Microsoft-Windows-Security-SPP	Application	9/21/2026 10:41:34 PM

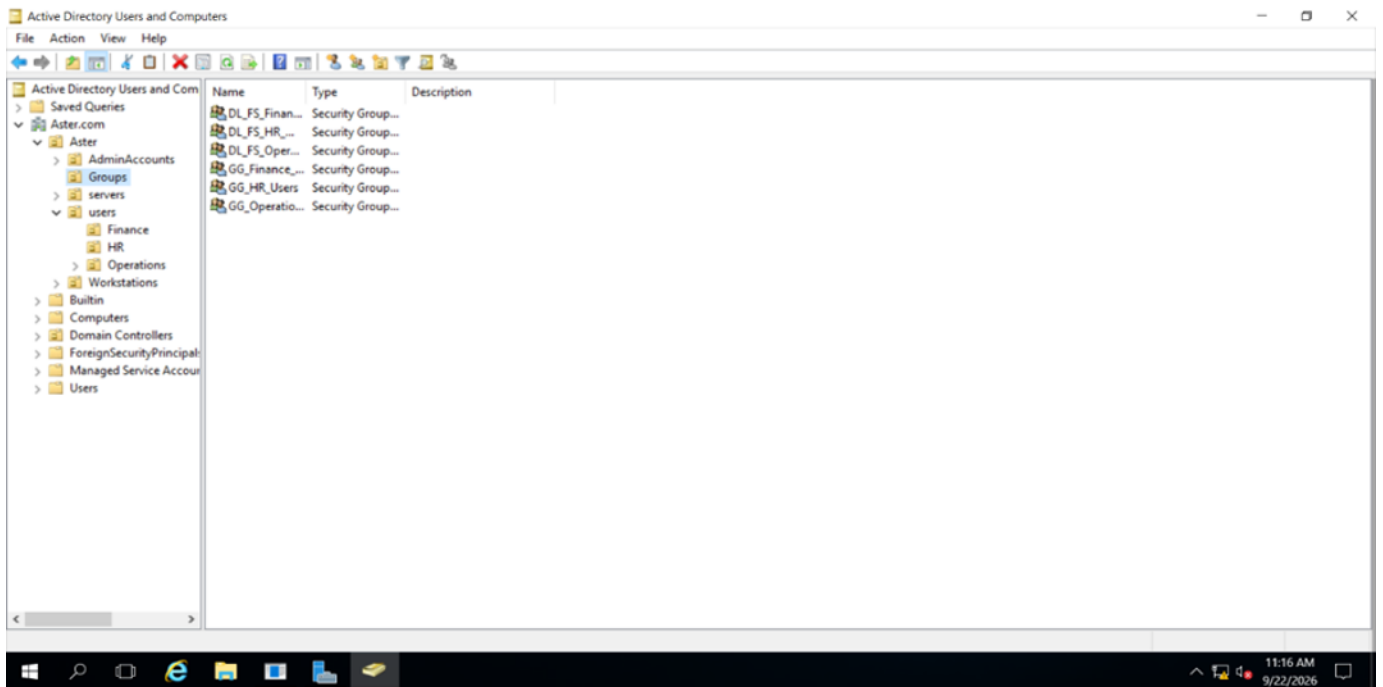
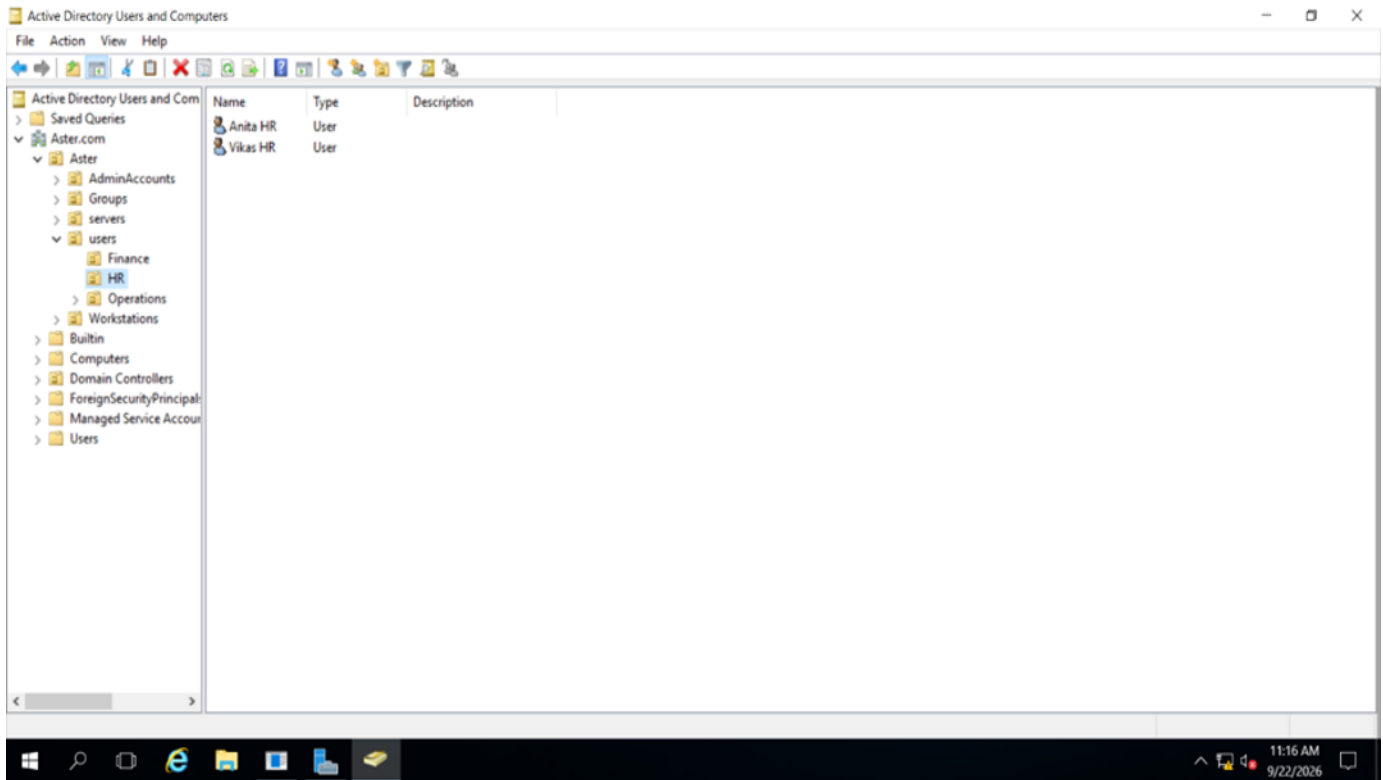
10:43 PM
9/21/2026

AST-BK01 Server Manager.

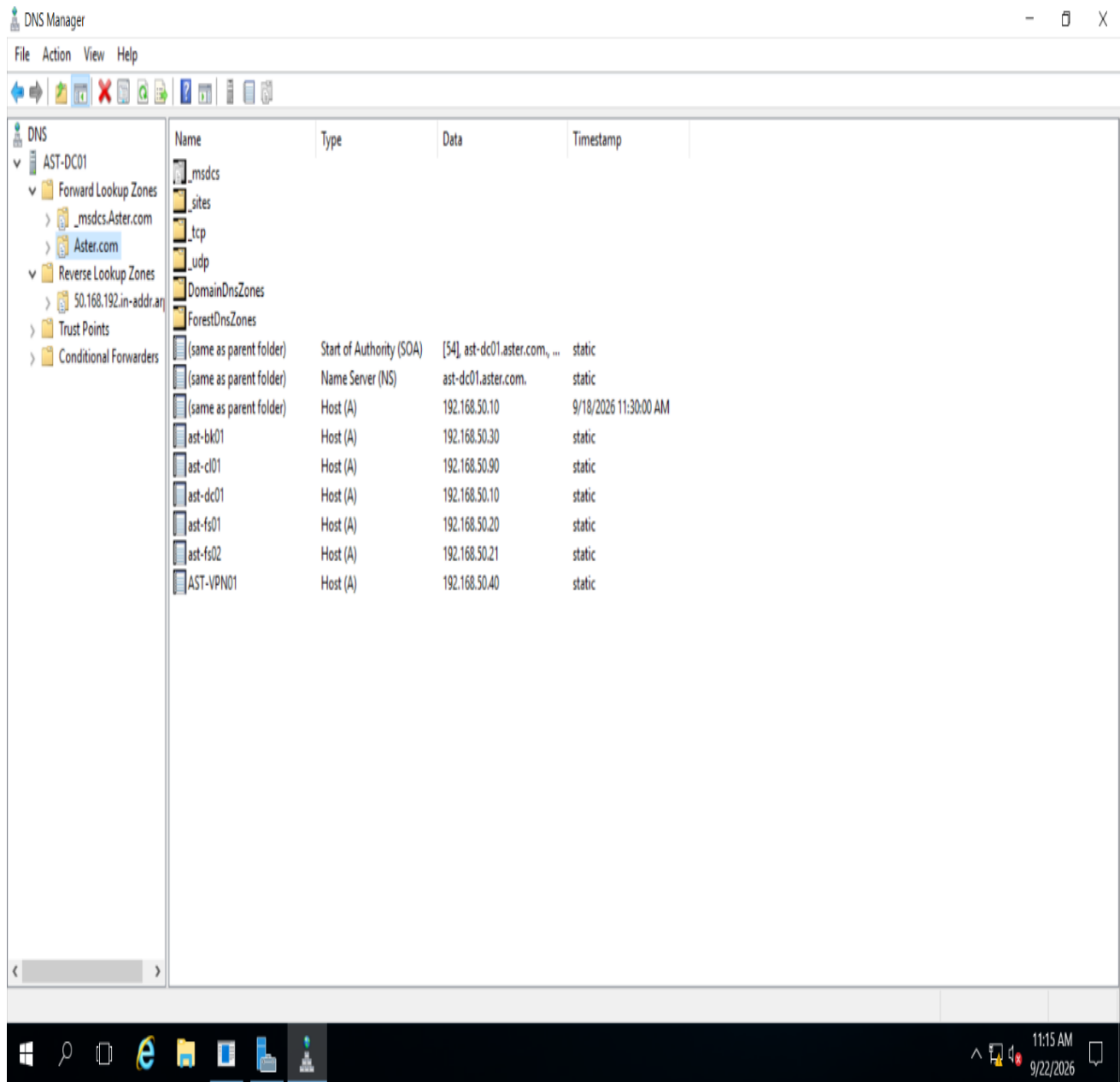
B05 — Build AST-CL01



B06 — Install Active Directory Domain Services

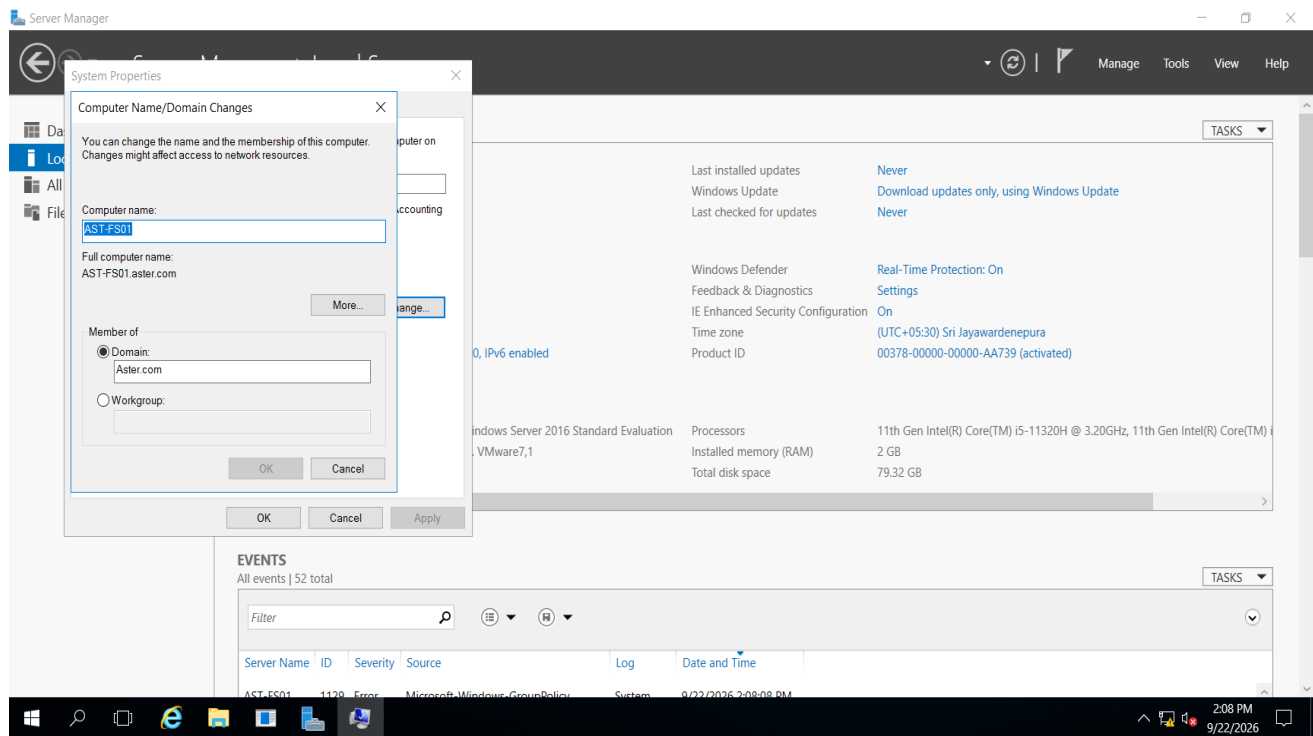


B07 — Create the forest and DNS service



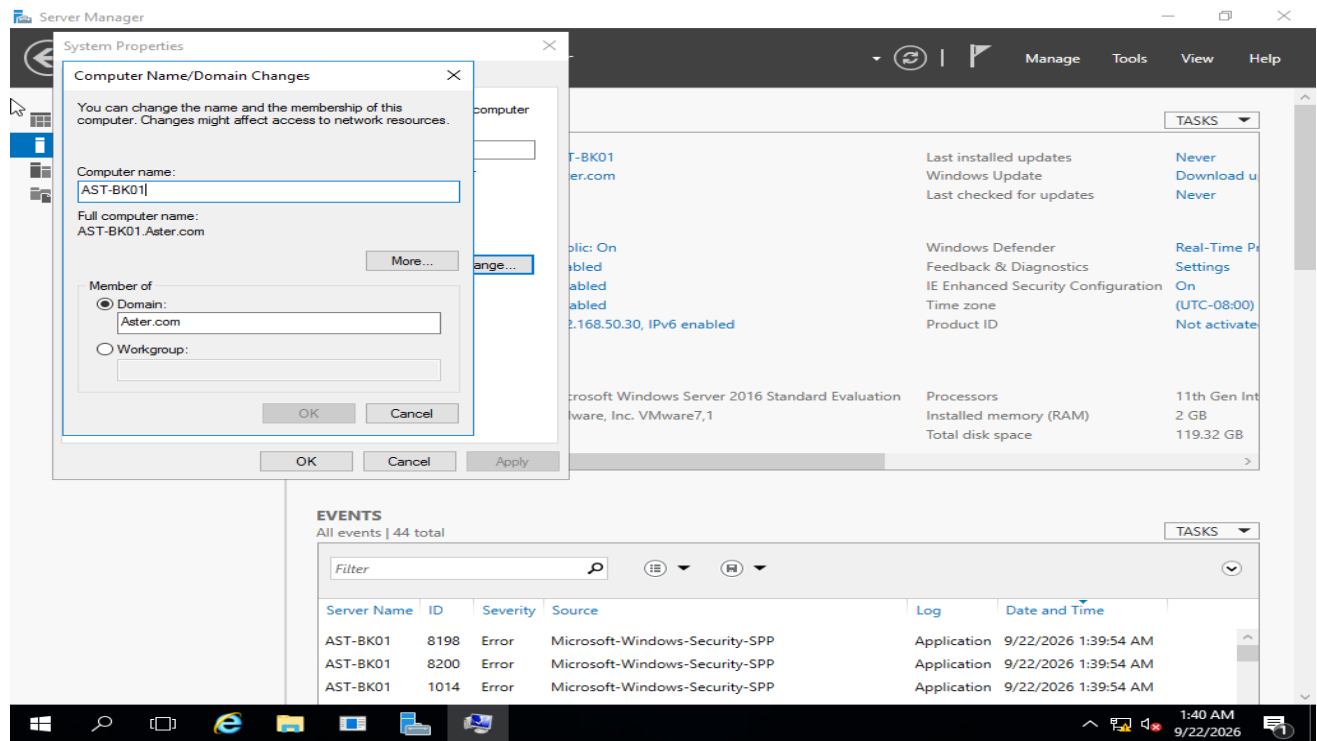
DNS Manager showing the Aster DNS zones and server records.

B08 — Join AST-FS01 to the domain



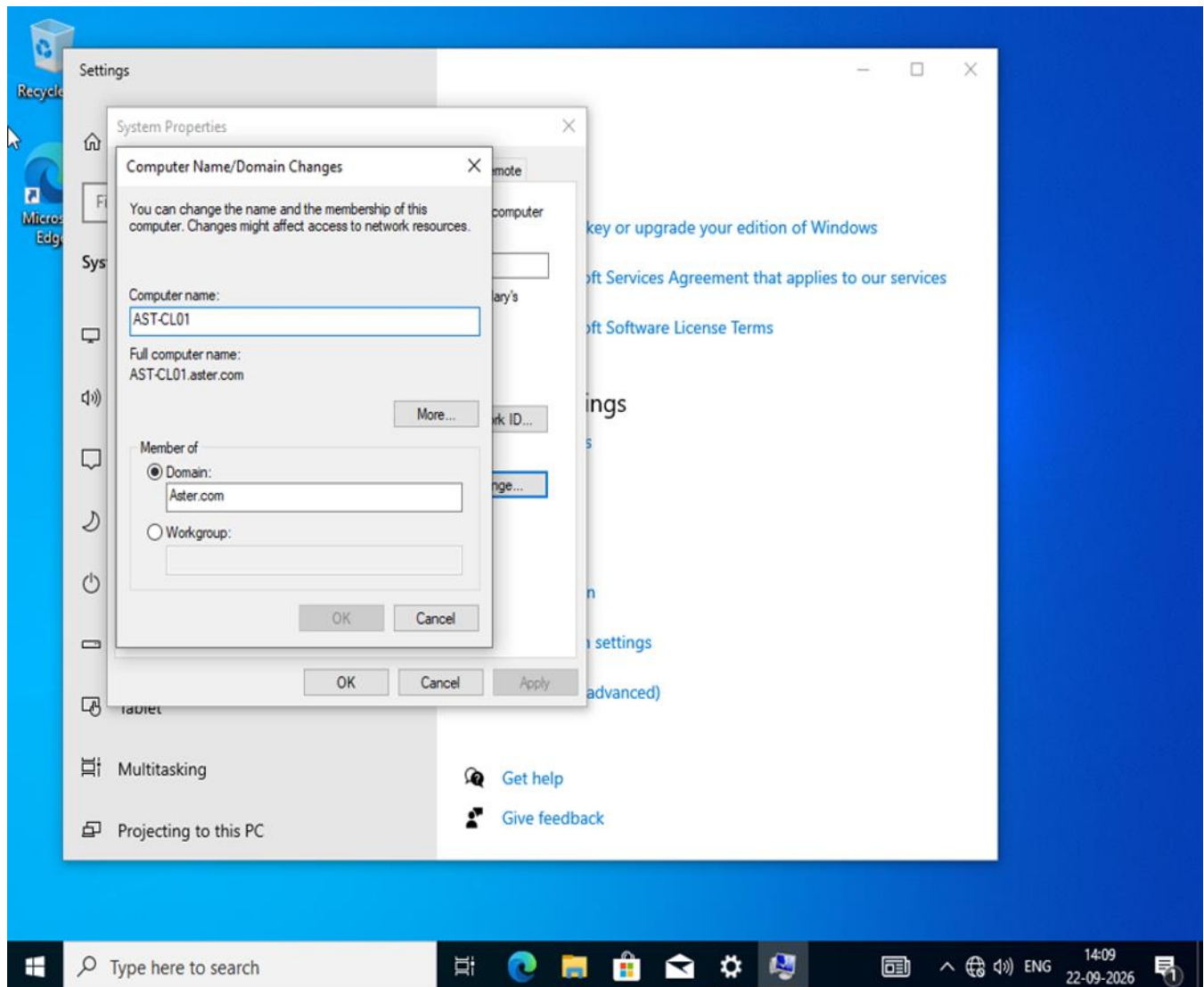
AST-FS01 Computer Name/Domain Changes showing membership in Aster.com.

B09 — Join AST-BK01 to the domain

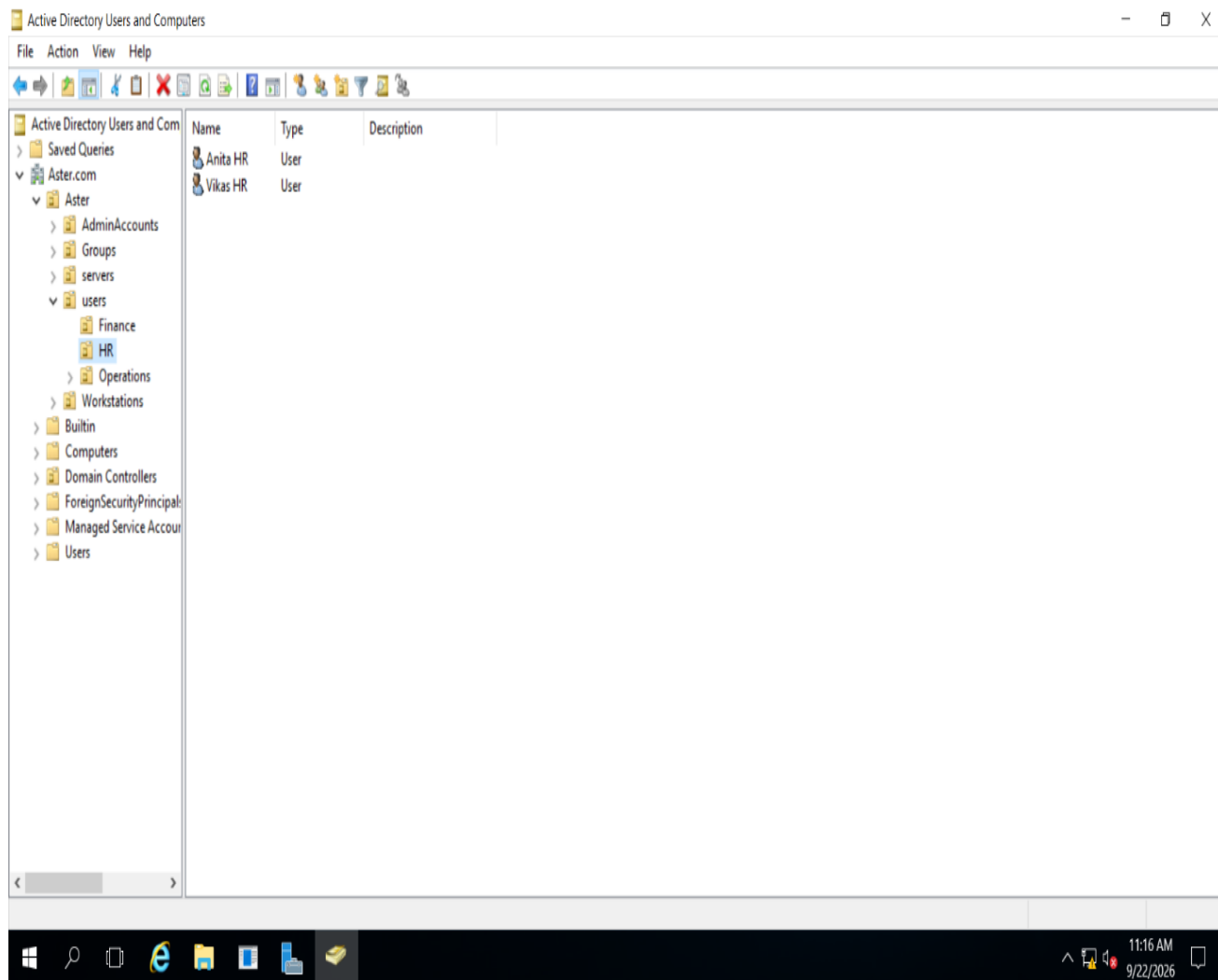


AST-BK01 Computer Name/Domain Changes showing Aster.com domain membership.

B10 — Join AST-CL01 to the domain

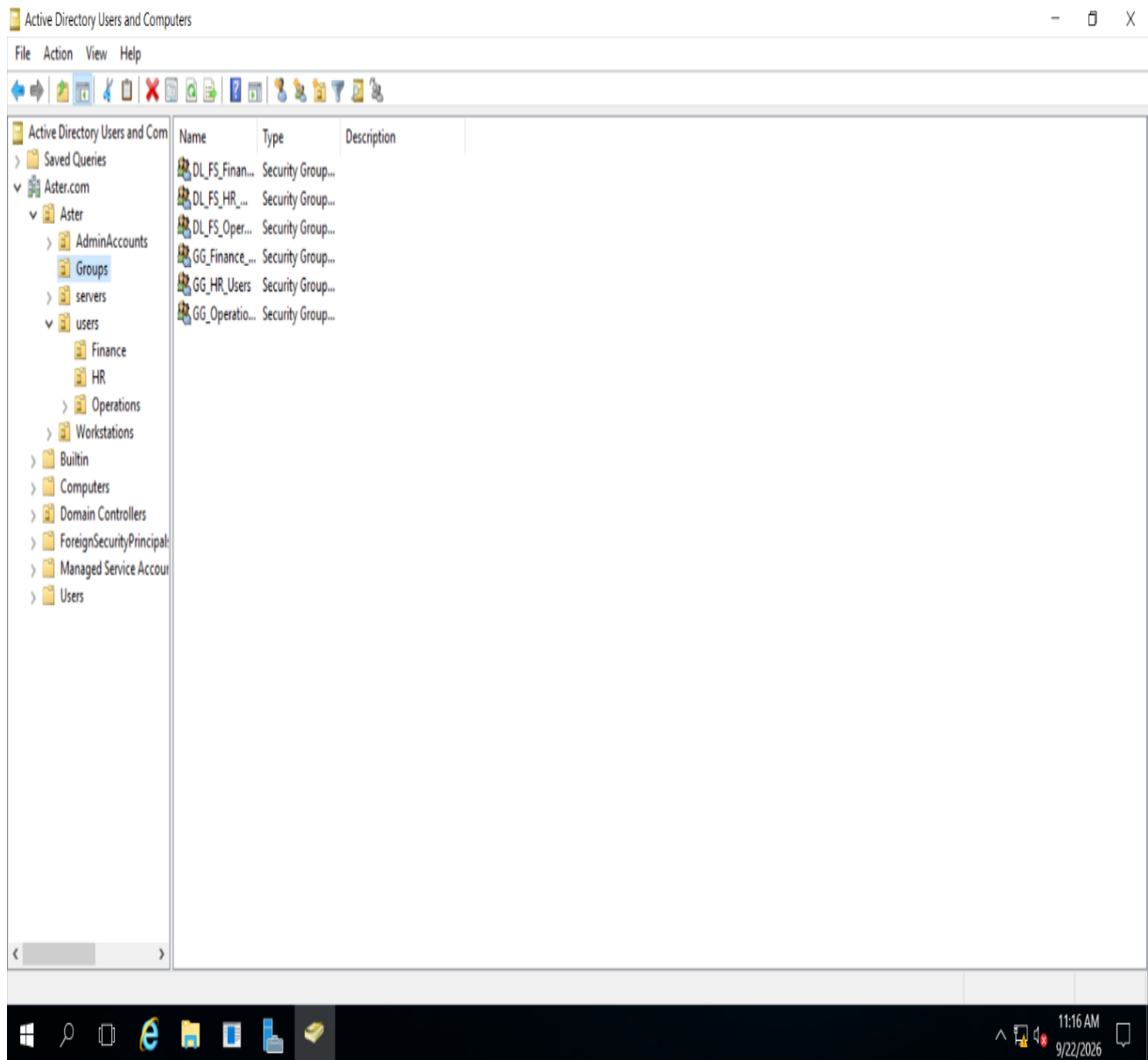


B11 — Create the organization and named accounts



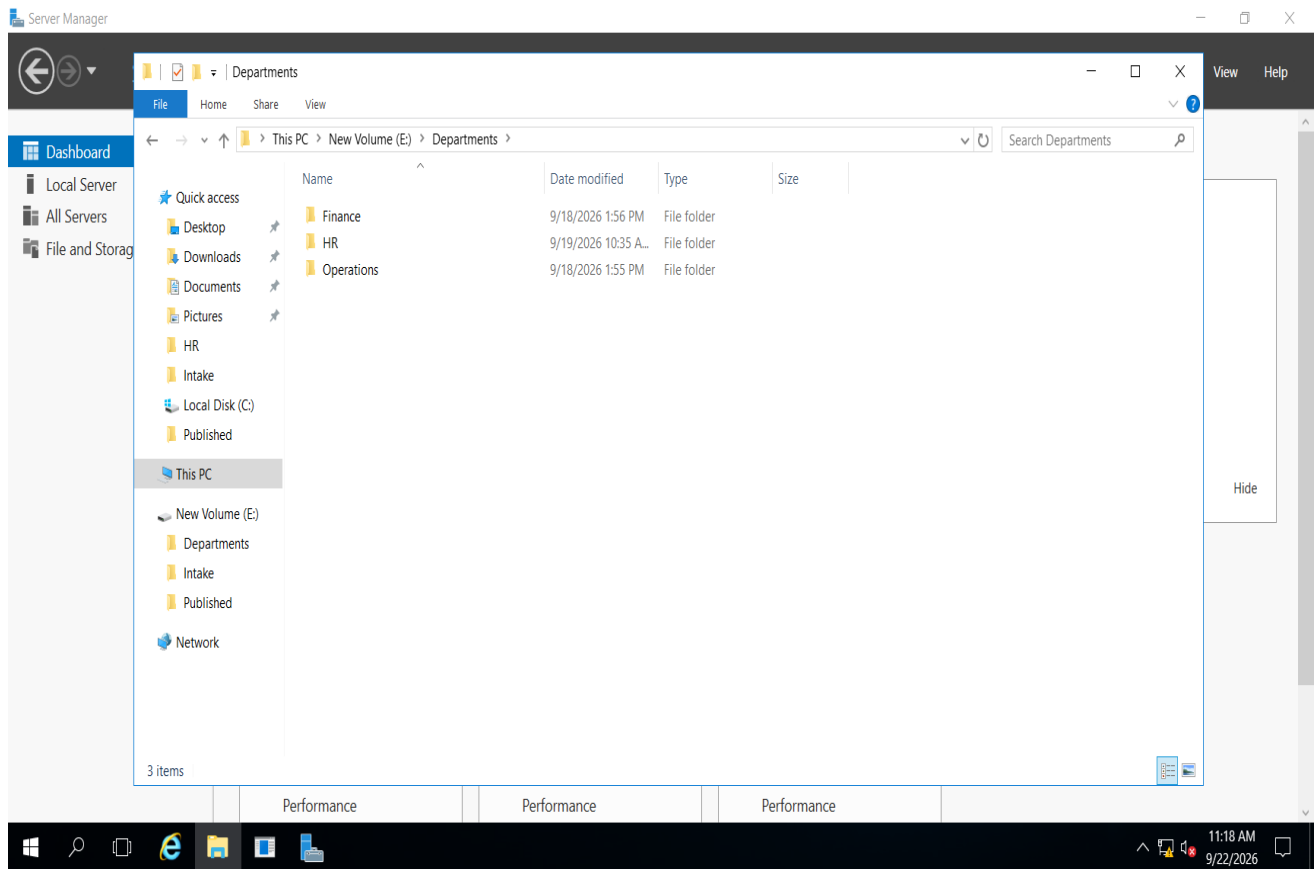
Active Directory Users and Computers showing named users and departmental OUs.

B12 — Build group based access assignments

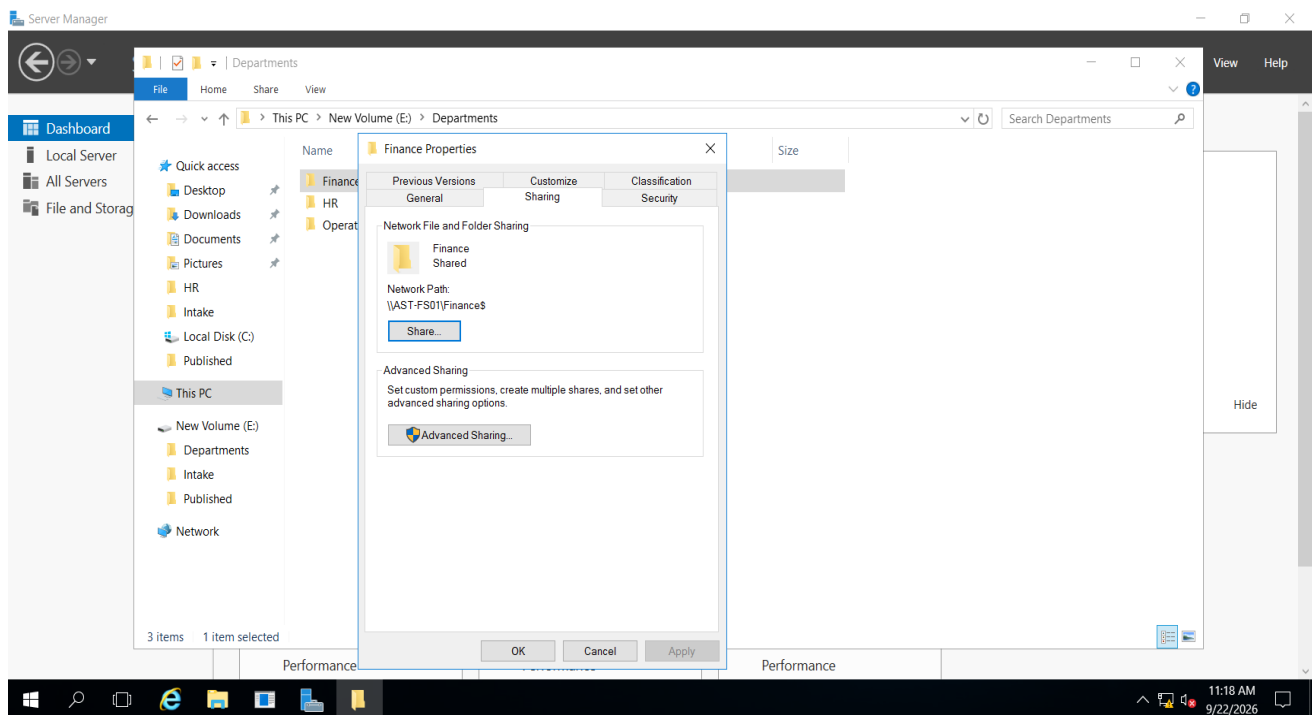


Active Directory Users and Computers showing Finance, HR, Operations and role groups.

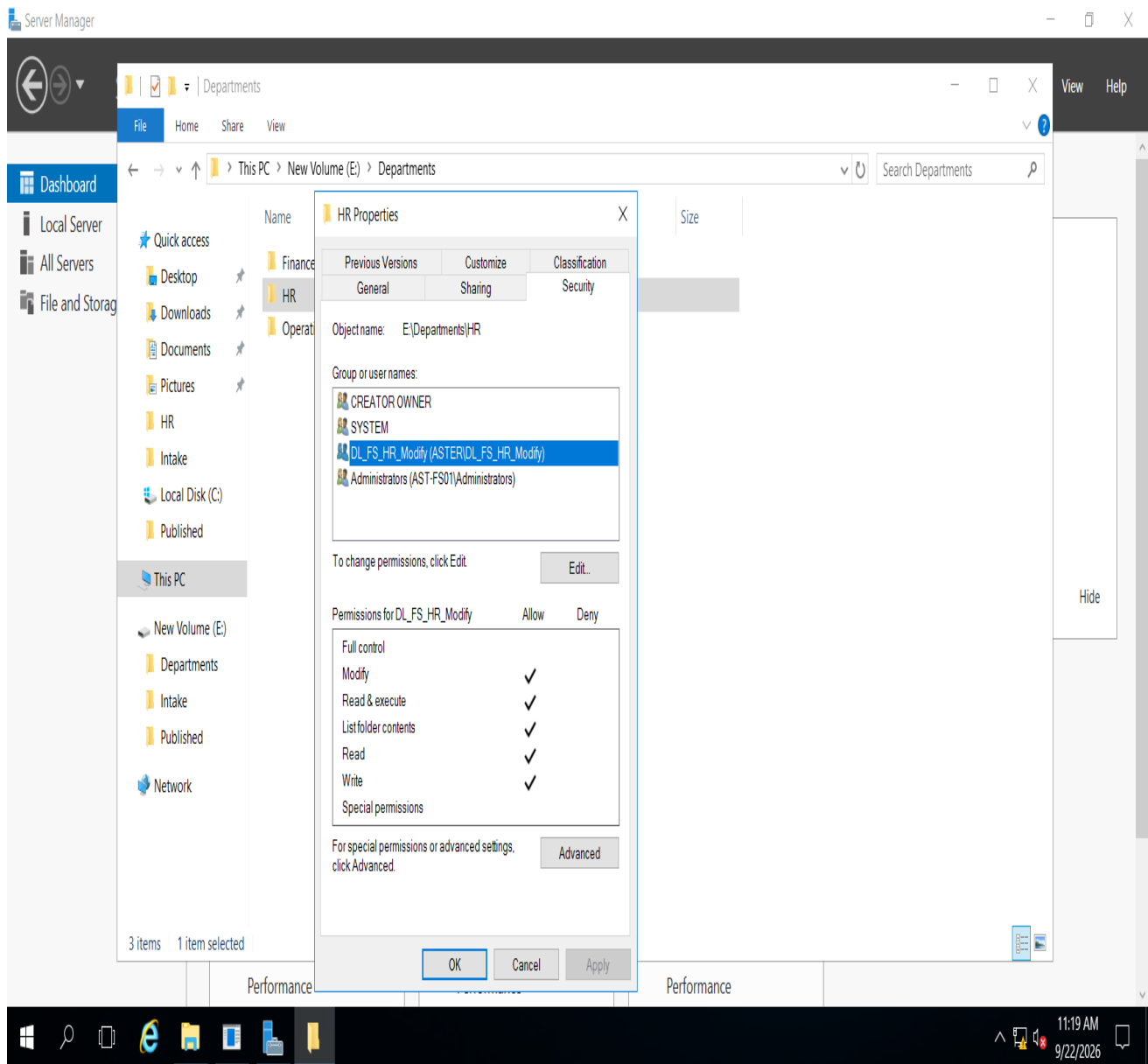
B13 — Publish three controlled departmental shares



Department folders showing Finance, HR and Operations.

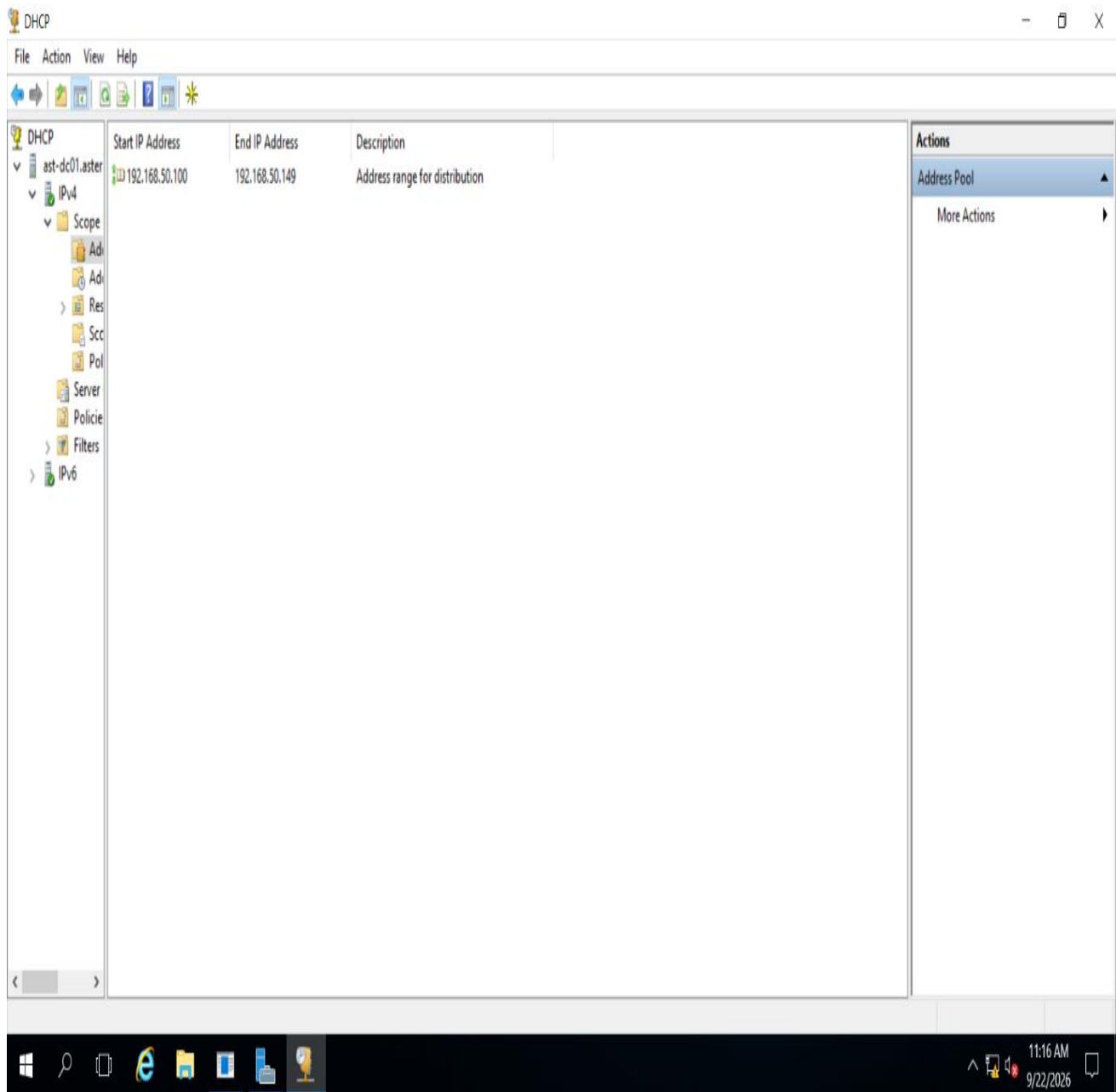


Finance share properties and share path.



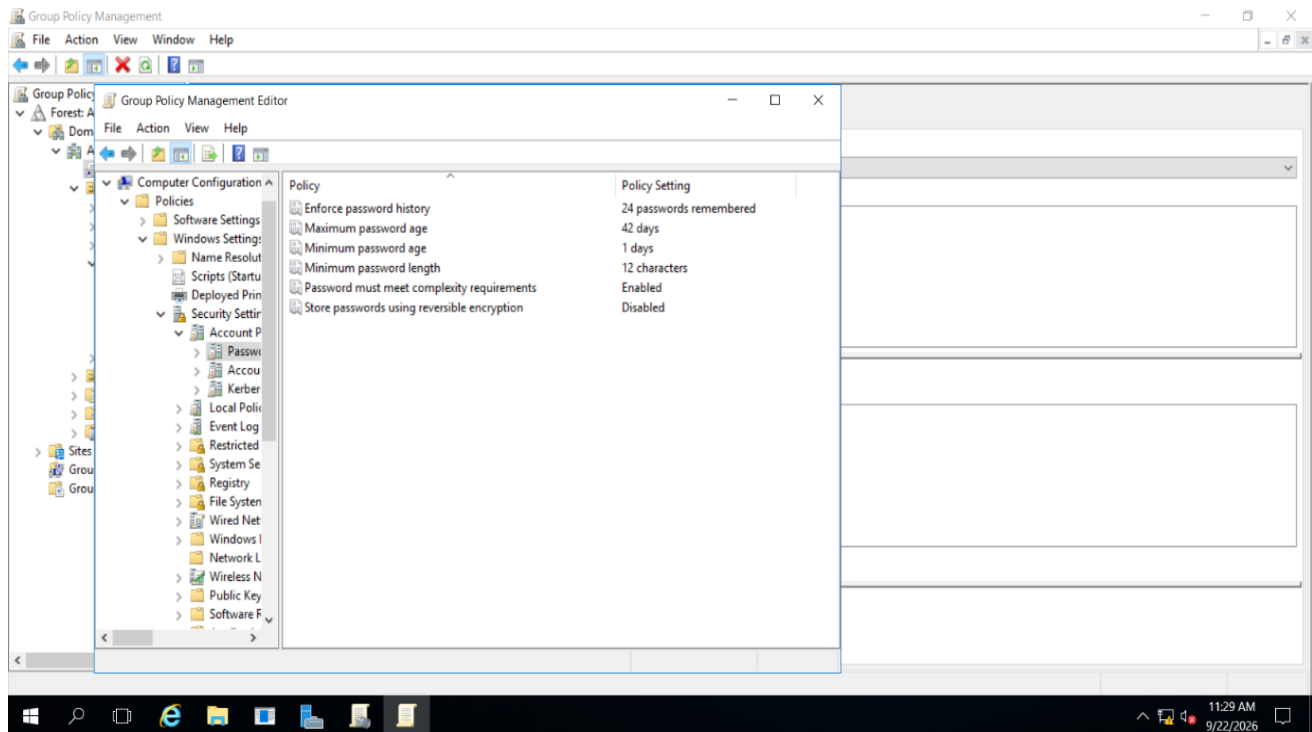
HR share security/permissions evidence.

B14 — Install and configure DHCP

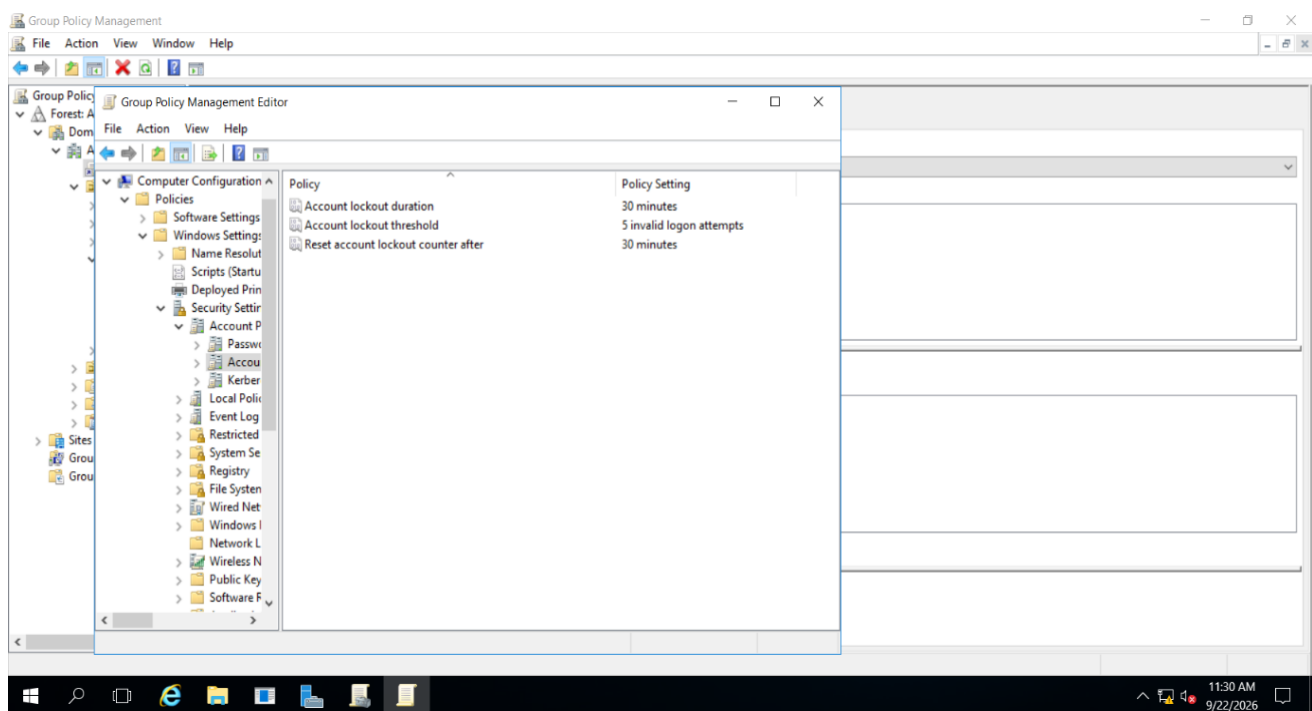


DHCP console showing the configured start and end address range.

B15 — Set domain account policy and HR user policy



Group Policy Management Editor showing password policy settings.



Group Policy Management Editor showing account-lockout settings.

P3-01 — Create the WAN network and two VPN lab guests

AST-VPN01

▶ Power on this virtual machine

▶ Edit virtual machine settings

▼ Devices

Memory	2 GB
Processors	2
Hard Disk (NVMe)	30 GB
CD/DVD (SATA)	Using file C:\Use...
Network Adapter	Custom (VMnet2)
Network Adapter 2	Custom (VMnet3)
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

▼ Description

Type here to enter a description of this virtual machine.

▼ Virtual Machine Details

State:

Powered off

Configuration file:

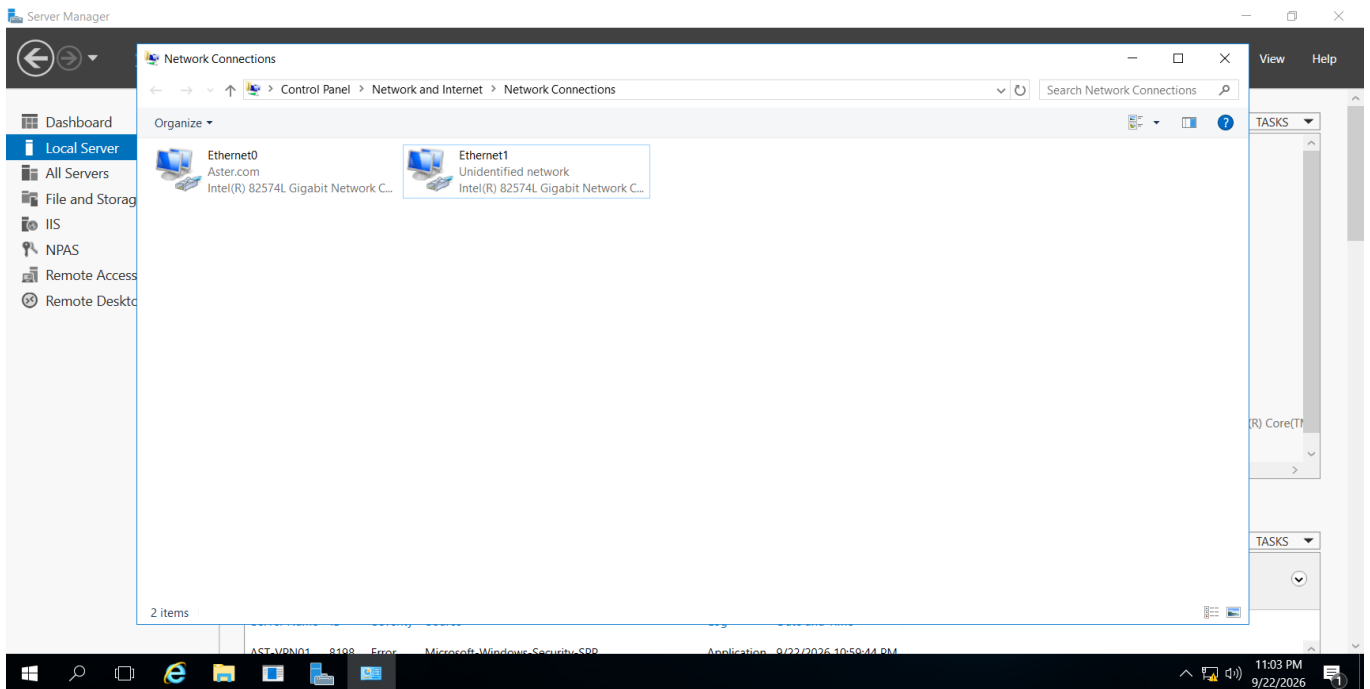
D:\MCSA\server5\Windows Server 2016.vmx

Hardware compatibility:

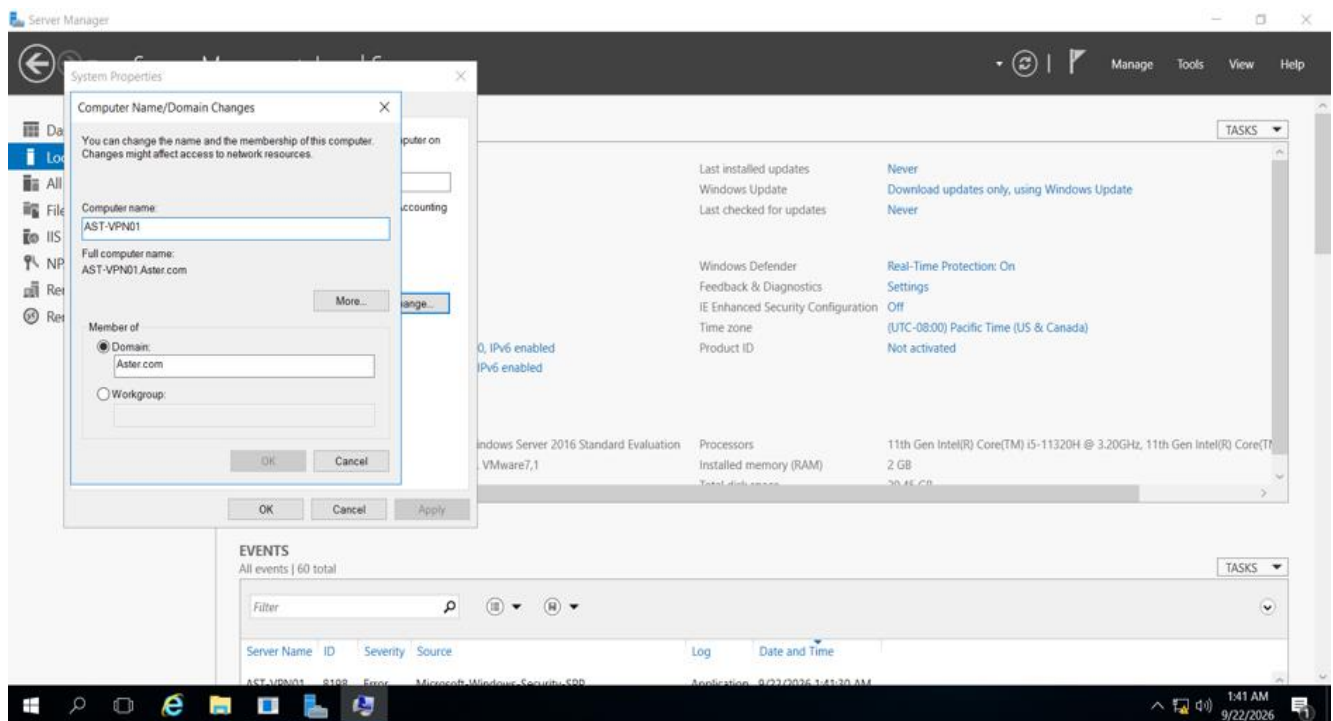
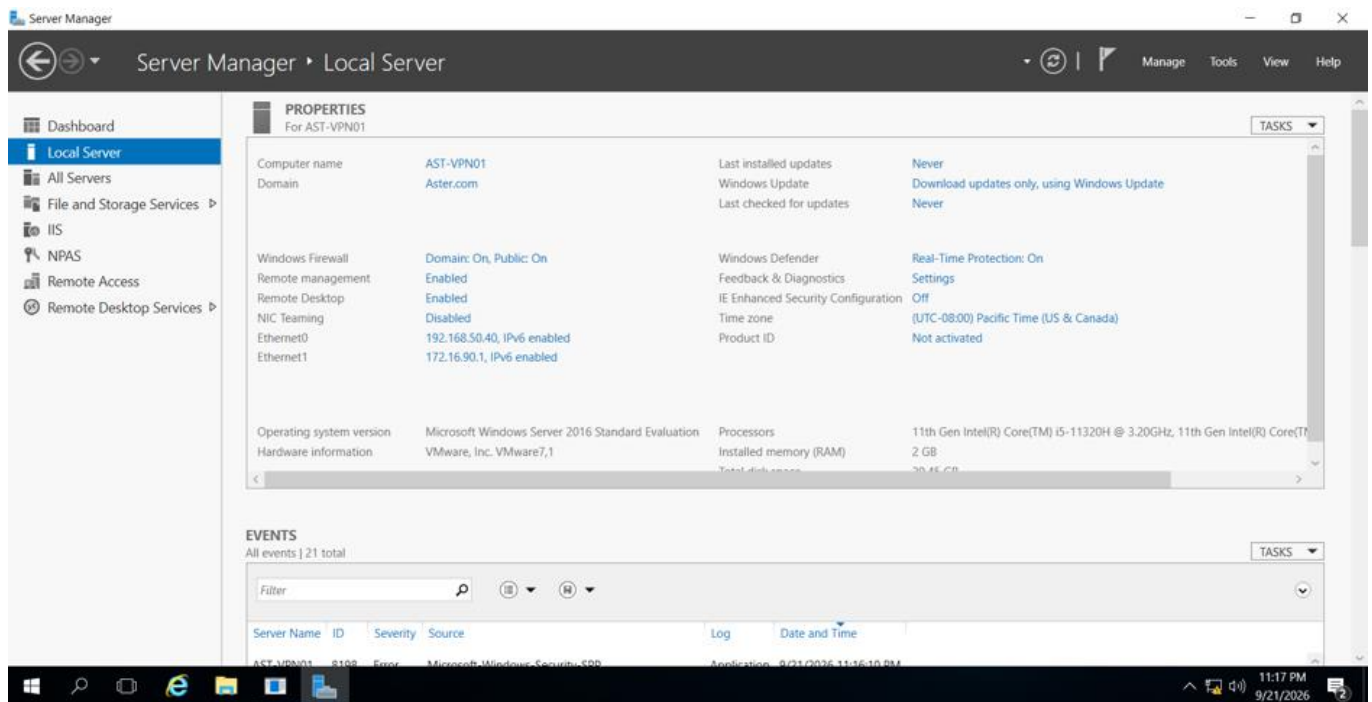
Workstation 16.2.x virtual machine

Primary IP address:

Network information is not available



AST-VPN01 Server Manager showing the VPN server, LAN 192.168.50.40 and WAN 172.16.90.1 interfaces.



```
Command Prompt
Microsoft Windows [Version 10.0.19045.3801]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Admin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

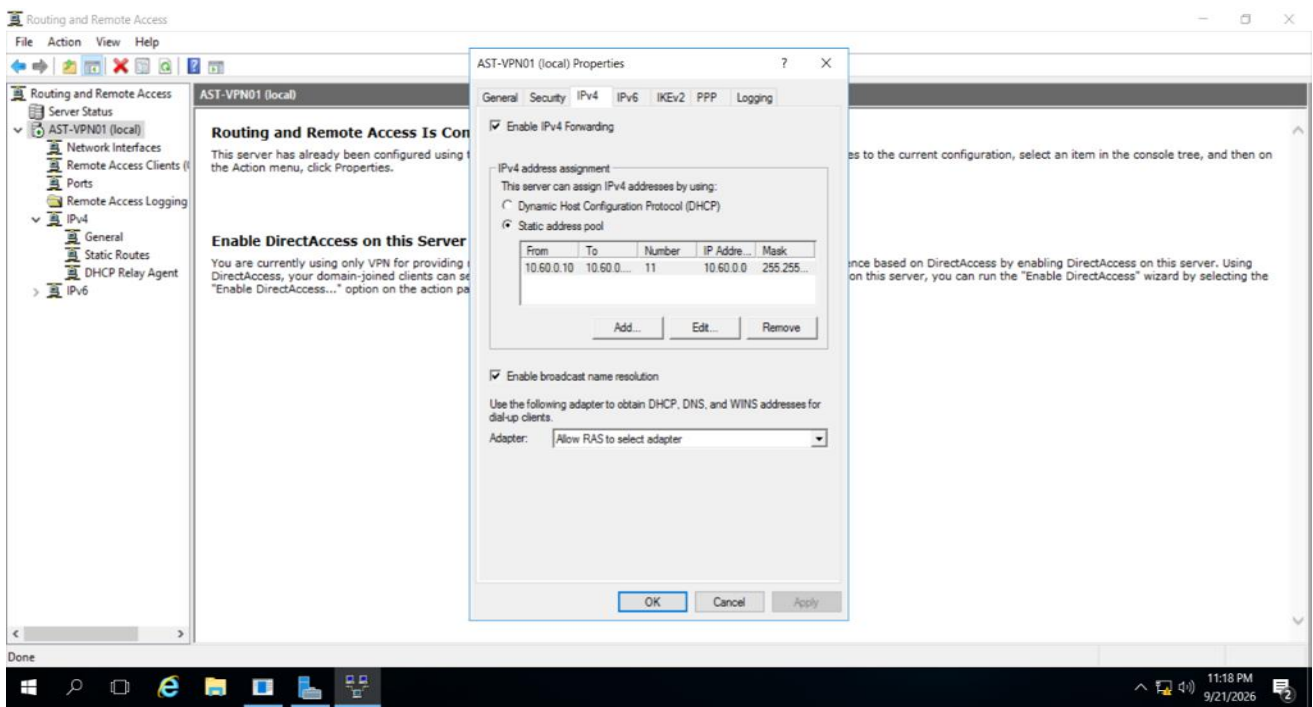
    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::57e8:2c58:655a:9ea9%14
    IPv4 Address. . . . . : 172.16.90.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

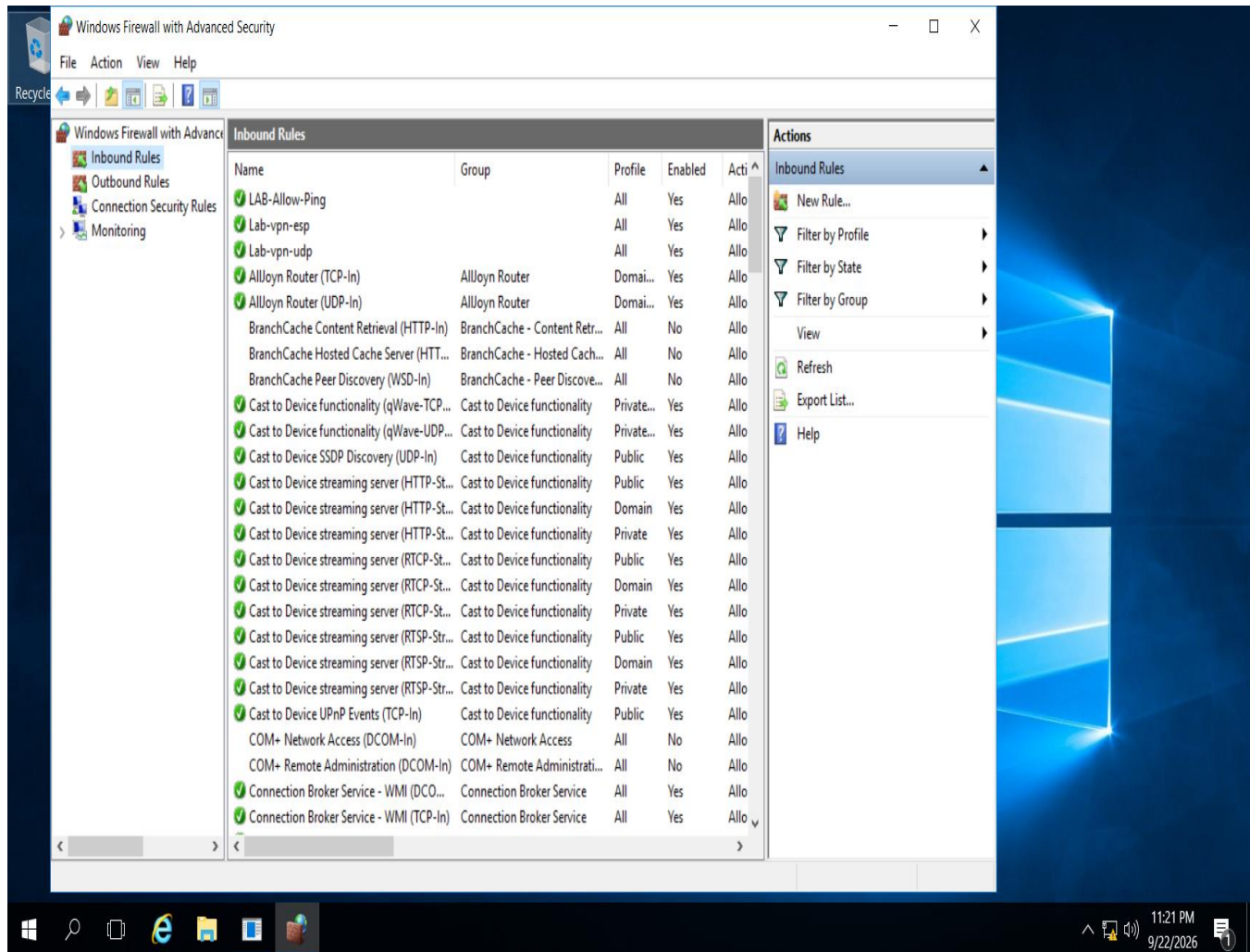
C:\Users\Admin>
```

P5-02 — Name interfaces and configure RRAS



RRAS IPv4 configuration supporting the VPN client address pool and forwarding.

P5-03 — Configure return routes and scoped firewall access



P5-04 — Authorise one remote user and configure the client

```
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ipconfig

Windows IP Configuration

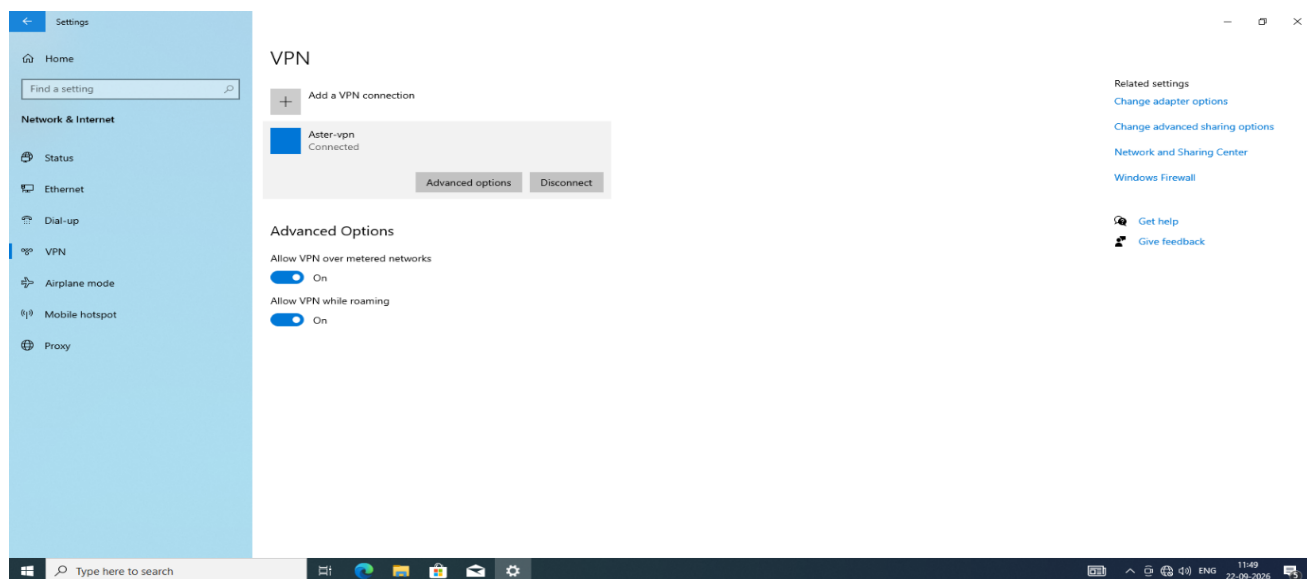
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::57e0:2c50:655a:9ea9%14
    IPv4 Address. . . . . : 172.16.90.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter Bluetooth Network Connection:

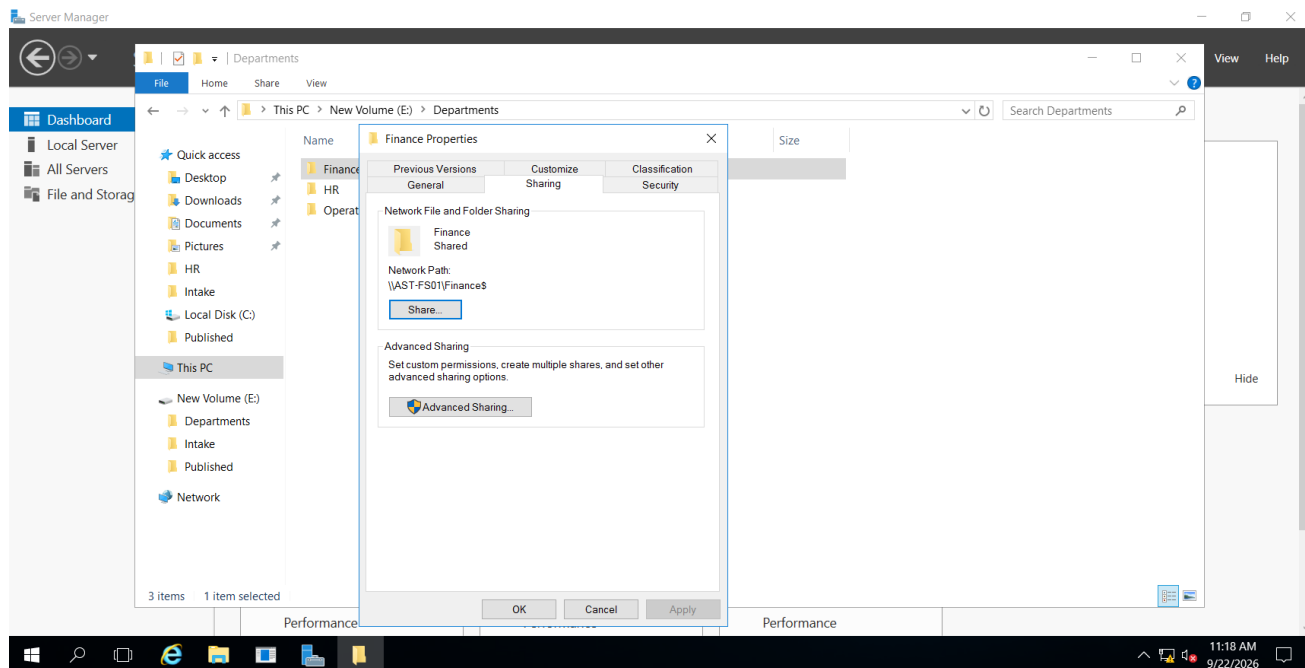
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\Admin>
```

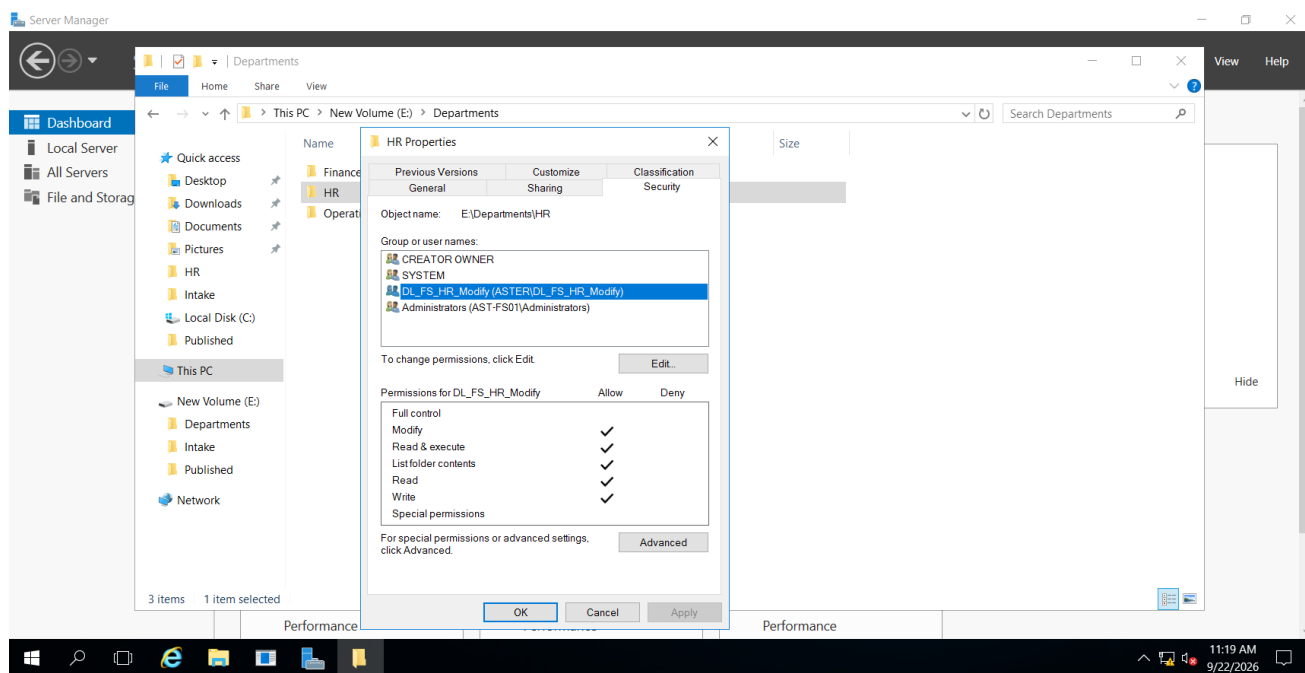


Windows VPN settings showing the Aster-vpn connection as Connected.

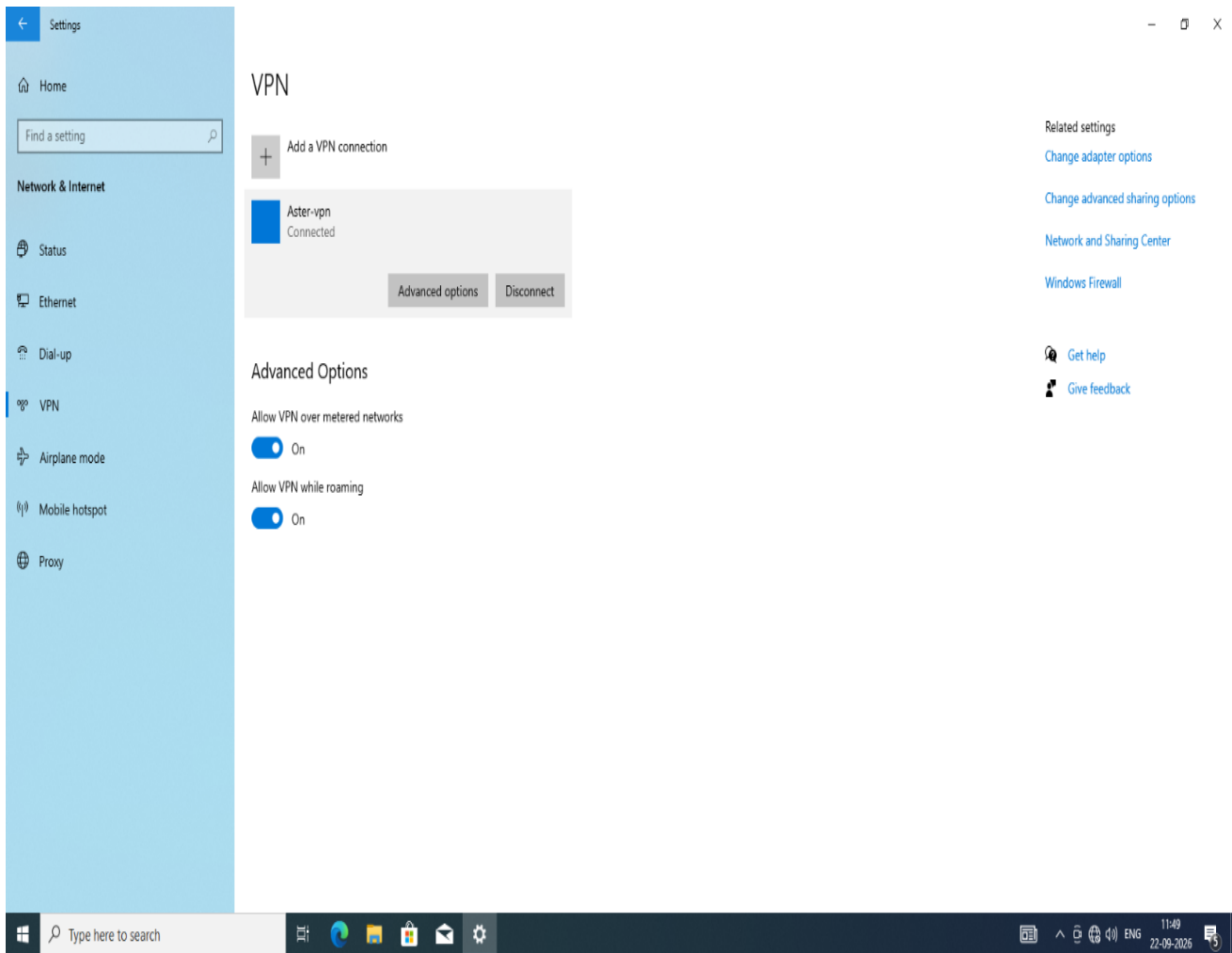
P5-05 — Prove DNS and share access through the tunnel



Finance share evidence used for the controlled-share permission baseline.

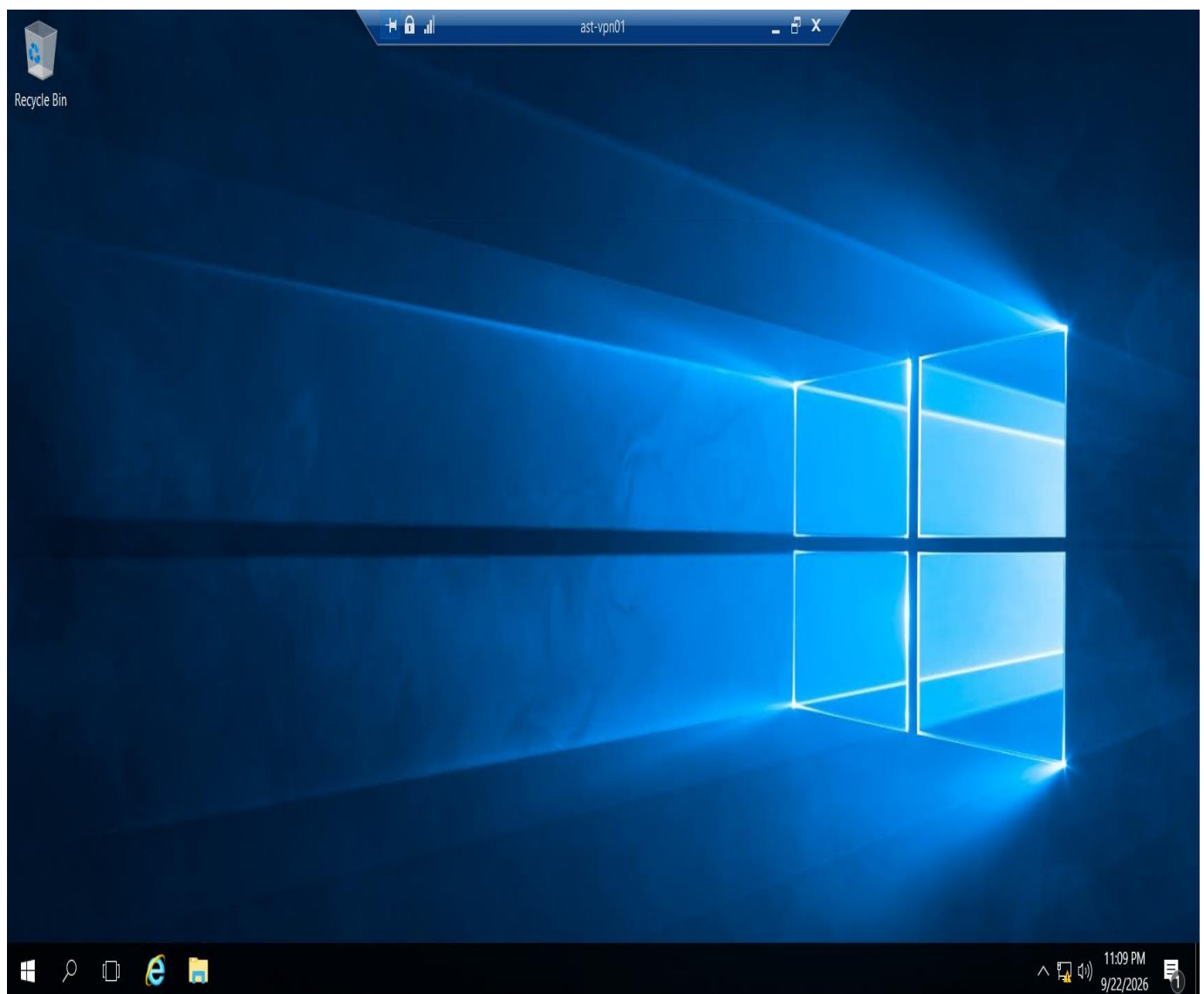


HR share security evidence used for the HR access baseline.



Connected VPN state on the remote client.

P5-06 — Use RDP through VPN and revoke remote access



Project Summary

This evidence report documents the Project 5 Remote Employee Access lab using the supplied Project 5 workbook and screenshot collection. The foundation build covers the Windows Server environment, Active Directory, DNS, DHCP, SMB/NTFS permissions and Group Policy. The specialist phase adds RRAS remote access, L2TP/IPsec, routing, dial-in authorization, VPN DNS/resource access, RDP with NLA and access revocation.

Key Skills Demonstrated

- VMware virtual network and virtual-machine administration
- Windows Server 2016 administration
- Active Directory and domain membership
- DNS and DHCP configuration
- SMB share and NTFS permission management
- Group Policy security configuration
- RRAS remote-access VPN
- L2TP over IPsec and VPN client configuration
- Routing and return-path configuration
- Windows Firewall scoped access
- RDP with Network Level Authentication
- Remote-access authorization and revocation
- Troubleshooting and validation

Evidence Completion Notes

The supplied screenshot document provides strong visual evidence for the foundation configuration and for the VPN server/client configuration. Dedicated screenshots for the P5-03 firewall rules, P5-05 DNS/SMB-over-VPN test, and P5-06 RDP/revocation test were not identifiable, so this report does not claim those screenshots exist.